

A Sample Article Title

Name1 Surname1¹ , Name2 Surname2 and Name3 Surname3

Abstract. The abstract should summarize the contents of the paper. It should be clear, descriptive, self-explanatory and not longer than 200 words. It should also be suitable for publication in abstracting services. Formulas should be used as sparingly as possible within the abstract. The abstract should not make reference to results, bibliography or formulas in the body of the paper—it should be self-contained.

Key words and phrases: first keyword, second keyword

INTRODUCTION

This template helps you to create a properly formatted manuscript for *Statistical Science*. Prepare your paper in the same style as used in this sample. Try to avoid excessive use of italics and bold face.

SECTION HEADINGS

Here are some sub-sections:

A Sub-section

Regular text.

A sub-sub-section. Regular text.

TEXT

Lists

The following is an example of an *itemized* list, two levels deep.

- This is the first item of an itemized list. Each item in the list is marked with a “tick.” The document style determines what kind of tick mark is used.
- This is the second item of the list. It contains another list nested inside it.
 - This is the first item of an itemized list that is nested within the itemized list.
 - This is the second item of the inner list.This is the rest of the second item of the outer list.
- This is the third item of the list.

The following is an example of an *enumerated* list.

1. This is the first item of an enumerated list.
2. This is the second item of an enumerated list.

Name1 Surname1, Department, University or Company Name, City, Country (e-mail: first@somewhere.com). Name2 Surname2, Department, Another University, City, Country (e-mail: second@somewhere.com). Name3 Surname3, Department, Third University, City, Country (e-mail: third@somewhere.com).

¹Some comment

Punctuation

Dashes come in three sizes: a hyphen, an intra-word dash like “*U*-statistics” or “the time-homogeneous model”; a medium dash (also called an “en-dash”) for number ranges or between two equal entities like “1–2” or “Cauchy–Schwarz inequality”; and a punctuation dash (also called an “em-dash”) in place of a comma, semicolon, colon or parentheses—like this.

Generating an ellipsis . . . with the right spacing around the periods requires a special command.

Citation

Only include in the reference list entries for which there are text citations, and make sure all citations are included in the reference list. Simple cite: **(author?)** [1]. Multiple bibliography items cite: [1–4]. Citing bibliography with object: **(author?)** [1], Theorem 1.

FONTS

Please use text fonts in text mode, e.g.:

- Roman
- *Italic*
- **Bold**
- SMALL CAPS
- Typewriter

Please use mathematical fonts in mathematical mode, e.g.:

- ABCabc123
- *ABCabc123*
- **ABCabc123**
- ***ABCabc123*** $\alpha\beta\gamma$
- $\mathcal{ABC}$
- $\mathbb{ABC}$
- ABCabc123
- ABCabc123
- $\mathfrak{ABCabc123}$

Note that `\mathcal` and `\mathbb` belong to capital letters-only font typefaces.

NOTES

Footnotes¹ pose no problem.²

ENVIRONMENTS

Examples for Plain-Style Environments

Theorem environments are defined in `header-includes` and used as raw LaTeX blocks. This is required because the IMS class patches `\theoremstyle` in ways that are incompatible with Quarto’s automatic theorem injection.

AXIOM 1. *This is the body of Axiom 1.*

PROOF. This is the body of the proof of the axiom above. □

CLAIM 2. *This is the body of Claim 2. Claim 2 is numbered after Axiom 1 because we used [axiom] in \newtheorem.*

¹This is an example of a footnote.

²Note that footnote number is after punctuation.

THEOREM 0.1. *This is the body of Theorem 0.1. Theorem 0.1 numbering is dependent on section because we used [section] after \newtheorem.*

THEOREM 0.2 (Title of the theorem). *This is the body of Theorem 0.2. Theorem 0.2 has additional title.*

LEMMA 0.3. *This is the body of Lemma 0.3. Lemma 0.3 is numbered after Theorem 0.2 because we used [theorem] in \newtheorem.*

PROOF OF LEMMA 0.3. This is the body of the proof of Lemma 0.3. □

Examples for Definition-Style Environments

DEFINITION 0.4. This is the body of Definition 0.4. Definition 0.4 is numbered after Lemma 0.3 because we used [theorem] in \newtheorem.

EXAMPLE. This is the body of the example. Example is unnumbered because we used \newtheorem* instead of \newtheorem.

FACT. This is the body of the fact. Fact is unnumbered because we used \newtheorem* instead of \newtheorem.

TABLES AND FIGURES

Sample of cross-reference to figure. Figure 1 shows the estimated density of the standard normal distribution.

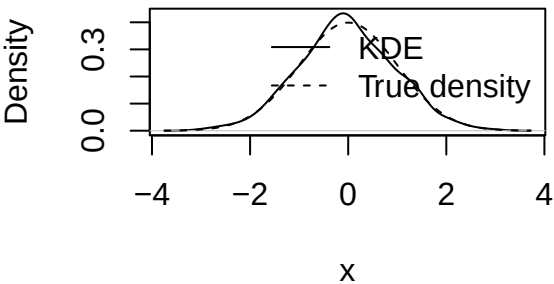


Figure 1: Kernel density estimate of 500 standard normal samples.

Cross-references to labeled tables: As you can see in Table 1 and also in Table 2.

Table 1: The spherical case ($I_1 = 0, I_2 = 0$)

Equil. points	x	y	z	C	S
L_1	\$-\$2.485252	0.000000	0.017101	8.230712	U
L_2	0.000000	0.000000	3.068884	0.000000	S
L_3	0.009869	0.000000	4.756387	\$-\$0.000058	U
L_4	0.210590	0.000000	\$-\$0.007021	9.440511	U
L_5	0.455927	0.000000	\$-\$0.212447	7.586127	U

Table 2: Sample posterior estimates for each model

Model	Parameter	Mean	Std. dev	2.5%	50%	97.5%
Model 0	β_0	\$-\$12.29	2.29	\$-\$18.04	\$-\$11.99	\$-\$8.56
	β_1	0.10	0.07	\$-\$0.05	0.10	0.26
	β_2	0.01	0.09	\$-\$0.22	0.02	0.16
Model 1	β_0	\$-\$4.58	3.04	\$-\$11.00	\$-\$4.44	1.06
	β_1	0.79	0.21	0.38	0.78	1.20
	β_2	\$-\$0.28	0.10	\$-\$0.48	\$-\$0.28	\$-\$0.07

EQUATIONS AND THE LIKE

Two equations:

$$(1) \quad C_s = K_M \frac{\mu/\mu_x}{1 - \mu/\mu_x}$$

and

$$G = \frac{P_{\text{opt}} - P_{\text{ref}}}{P_{\text{ref}}} 100(\%).$$

Equation arrays:

$$\frac{dS}{dt} = -\sigma X + s_F F,$$

$$\frac{dX}{dt} = \mu X,$$

$$\frac{dP}{dt} = \pi X - k_h P,$$

$$\frac{dV}{dt} = F.$$

One long equation:

$$\begin{aligned} \mu_{\text{normal}} &= \mu_x \frac{C_s}{K_x C_x + C_s} \\ &= \mu_{\text{normal}} - Y_{x/s} (1 - H(C_s)) (m_s + \pi/Y_{p/s}) \\ &= \mu_{\text{normal}}/Y_{x/s} + H(C_s) (m_s + \pi/Y_{p/s}). \end{aligned}$$

APPENDIX

Appendices should be provided in the Appendix section, before Acknowledgements. If there is only one appendix, please refer to it as the Appendix. If there are multiple appendices, refer to them as Appendix A, Appendix B, etc.

ACKNOWLEDGMENTS

The authors would like to thank the anonymous referees, an Associate Editor and the Editor for their constructive comments that improved the quality of this paper.

FUNDING

The first author was supported by NSF Grant DMS-00-000000.

The second author was supported in part by NIH Grant 000000000.

SUPPLEMENTARY MATERIAL

Title of Supplement A

Short description of Supplement A.

Title of Supplement B

Short description of Supplement B.

REFERENCES

- [1] BILLINGSLEY, P. (1999). *Convergence of Probability Measures*, 2 ed. Wiley, New York.
- [2] BOURBAKI, N. (1966). *General Topology* **1**. Addison-Wesley, Reading, MA.
- [3] ETHIER, S. N. and KURTZ, T. G. (1985). *Markov Processes: Characterization and Convergence*. Wiley, New York.
- [4] PROKHOROV, Y. V. (1956). Convergence of random processes and limit theorems in probability theory. *Theory of Probability & Its Applications* **1** 157–214.